

FOUNDATIONS OF OPERATIONAL RESEARCH AND ANALYTICS



Simulation Course Work

Ali Saeed Ltfallah

LtfallahAS@cardiff.ac.uk

**Using a validated QSR model to derive general principles for
operational improvement.**

1.1 Introduction:

The Quick-Service Restaurant (QSR) industry operates on razor-thin margins where operational efficiency is paramount to both profitability and customer retention. In recent years, the landscape has become increasingly complex with the rise of multiple ordering channels—including in-store, takeaway, and app-based delivery—each with unique demand patterns and service expectations. This creates a dynamic and challenging environment where managers must balance staffing levels, kitchen workflow, menu complexity, and physical space constraints to deliver a fast, high-quality experience.

Traditional static analysis tools, such as spreadsheets, often fail to capture the interdependent and stochastic nature of these operations. A bottleneck at a single cooking station, for instance, can create a cascading delay that impacts customer queues, food quality, and ultimately, the bottom line. This project addresses these challenges by developing a validated, discrete-event simulation model of a typical high-volume QSR. The goal is to create a "digital laboratory" to identify common operational bottlenecks and test the impact of various process improvement strategies.

To ensure the model's accuracy and relevance to real-world conditions of the QSRs, it was calibrated and validated using primary operational data obtained from a high-traffic KFC franchise in Cardiff, Queen Street, as well as other data sources related to QSR to enhance its output and ensure its accuracy. This method follows the simple philosophy of creating an accurate baseline model that represents the real world operations of the modern fast food restaurants, and once this has been achieved then trust can be gained in finding true insights.

The 3 Sections below follow the pages constraint as required by the coursework submission guidelines.

Section 1: Data Collection and Analysis

1.2 Objectives and Aims:

Using this validated model, this study will conduct a series of simulation experiments to answer key operational and strategic questions relevant to the QSR industry. The objectives are to move beyond a simple descriptive model and use the simulation as a predictive tool for data-driven decision-making.

1. To Identify and Quantify System Bottleneck: To determine the primary and "shifting" bottlenecks within the system—from order taking to kitchen production and final packing. The analysis will measure the impact of these constraints on critical KPIs such as overall throughput, customer wait times, and staff utilization.

2.To Conduct a Profitability and Menu Engineering Analysis:

To analyze the financial performance of the operation at a granular level. This includes:

- **Item Profitability:** Using the model's integrated financial layer to determine the profitability of individual menu items and order channels (Eat-In, Takeaway, Delivery).
- **Menu Optimization:** Evaluating the system-wide impact of strategic menu changes, such as moving a complex, low-margin item to a more efficient "ready-to-heat" process, and quantifying the resulting effect on both throughput and overall profitability.
- **Analyzing problems such as the "Cold fries" problem while waiting for a more complex item to be cooked in the same combo meal, and propose possible solutions.**

3. To Evaluate Staffing and Workflow Strategies:

To assess the effectiveness of different staffing models. This involves analyzing the impact of dynamic shift changes in response to fluctuating demand and quantifying the significant financial and operational benefits of cross-training kitchen staff to create a more flexible and responsive workforce.

4. To Analyze the Impact of Physical and Resource Constraints: To measure the financial cost of lost business due to physical and resource limitations. This includes:

- **Physical Space:** Quantifying the revenue lost from "Eat-In" customers who balk due to a lack of available seating.
- **Resource Competition in a Wider Market:** Transforming the model to an archetype to explore scenarios relevant to the broader QSR market (e.g., McDonald's and other competitors). This includes analyzing situations where arrivals might overwhelm the standard workflow.

5. To Propose and Test Novel Process Improvements:

To evaluate the system-wide impact of introducing new technologies and processes. This involves conducting "what-if" analysis on novel solutions such as an automated table allocation system and an app-based pre-ordering system, quantifying their potential return on investment and their effect on customer flow and satisfaction across all ordering channels.

2.0 Data Collection And Analysis:

In this section I will detail the methods in which I collected the data and informed the parameters of the model. I ensured to follow a standard professional approach that fits the purpose of the model.

2.1 Summary of Data Collection Methodology:

To ensure the simulation model accurately reflects the complex, real-world conditions of a high-volume QSR, a **triangulated data collection strategy was employed**. This methodology combines primary, observational and secondary data to create a robust and sound grounded foundation for the model's parameters, minimizing reliance on purely statistical assumptions and false generalizations. This approach involved three key stages:

A Source Data Collection (Survey): A structured survey was conducted with the manager of a representative KFC franchise. This provided critical internal operational parameters, such as the percentage split between ordering channels (Eat-In, Takeaway, Delivery), dynamic staffing schedules, and baseline kitchen processing time .

A Direct Observation: To capture the nuanced service times of the customer journey (such as time spent at the kiosk and time at the table), direct, timed observations were conducted. This observational methodology was guided by the approach outlined by Curin et al. (2005) in their study of a similar QSR environment, ensuring a systematic and repeatable process for data gathering.

A Secondary Data Analysis: Publicly available data from Google Maps was used to model macro-level demand patterns, providing the basis for a dynamic, Time Dependent Distribution that accurately reflects the daily ebb and flow of customer traffic, including lunch and evening peaks. This fits these specific parameters better than direct observation or surveys, since it provides a real time busy and normal hours patterns.

This hybrid approach ensures that all key model inputs are informed by a combination of high-level patterns, expert operational knowledge, and direct empirical evidence.

2.2 Source Data Collection (Survey): Initially, I conducted an online research to find relevant data about the restaurant being used to inform the model's parameters (KFC), while the results were satisfactory in the sense that they provided general information about the various arrivals channels (Delivery, Eat In, Takeaway), the data lacked a coherent statistical links that can be used to derive accurate parameters for the model.

Therefore, to solve this problem, I formulated a structured survey that was then conducted with the manager at a high-volume KFC franchise (Queen Street, Cardiff). **The survey** made sure to collect data that was otherwise difficult to collect by observing such dynamic staff levels, processing time for meals, back in the kitchen operations, bulking customers in a busy day and data for arrivals and priorities of the operations. **The survey was conducted on 7/11/2025 at 1:30 P.M, the restaurant was quite and this ensured a polite approach to the manager that does not disrupt the operations of the restaurant.**

Data Collection Parameter	Size
Percentage of Eat In	50%
Percentage Of Delivery	25%
Percentage Of Takeaway	25%
Processing time for a complex meal	0.5-1 Min
Processing time for a simple meal	0.5 Min
Number Of Cashiers(Front Staff)	Normal Hour : 2 - Busy Hour : 4
Number Of Cooks	Normal Hour : 1 - Busy Hour : 2
Number Of Packers(Packing Orders)	Normal Hour : 2 - Busy Hour : 3
Number Of Cleaners	Normal Hour : 0 Busy Hour : 1
Top 3 Selling Meals	1:Burger Meal, 2:Chicken Bucket, 3:Wings
Orders on average in a normal hour	25-30
Orders on average in a busy hour	70
Do cooks prepare in advance some meals	Yes
Time for canceling order if customer did not collect	5-25 minutes
Percentage Of Customer not picking up orders immediately	1-2%
Do you waste or reserve it to other customers if not picks up it up?	Waste
Number Of Customers Leaving if queue is long in a busy day	3-10 people
Does Delivery have priory over in store orders?	Yes

2.3.Direct Observational Study

To gather detailed data on customer service times, such as the average time spent ordering at a kiosk or a cashier and the average time a customer occupies a table, a direct observational study was conducted at a KFC branch(Cardiff). Following the methodology for collecting service time data outlined by Curin et al. (2005).

A stopwatch was used to calculate the typical customer journey of a 15 customers from arrival to leaving, of various operations.

The kiosk interaction times average is about 2 minutes, while table occupancy times vary from a brief 15 minutes to a longer 42-minute stay, time for collection does not exceed 10 seconds usually. Cashiers ranged from 2.5 minutes to 4 minutes.

2.4.Google map Data:

Google map was used to extract the popular times hourly graph for a typical weekday.

This data was used to create a Time Dependent Distribution. This allows the model's arrival rate to dynamically change throughout the day, realistically simulating the lunch and evening peaks.

Time Slots	Orders
10 AM - 15 PM	28 per hour /avg
15 PM - 19 PM	70 per hour /avg
19 PM - 00:00 AM	28 per hour /avg

2.5 Model Parameterization Table: The following table shows the methods in which the model was informed using the data above:

Parameter	Data Source(s)	SIMUL8 Implementation & Value	Rationale
Customer Arrivals (Volume & Mix)	Manager Survey (50/25/25 split) & Google Maps (Hourly patterns)	Three Start Points (Eat-In, Takeaway, Delivery), driven by a Time Dependent Distribution with some randomness.	To accurately model the fluctuating, multi-channel demand throughout a typical weekday, reflecting both real-world volume and customer choice.
Staffing Levels	Manager Survey (Specific staff counts for busy/normal hours)	Resources by shifts.	To reflect staffing operations in the restaurant.
Kitchen Processing	Manager Survey (0.5-1 min complex, 0.5 min simple)	Activities of the kitchen were sat using an average of the data provided by the survey.	To reflect the validated, high-efficiency "touch times" provided by the manager. Sensitivity analysis confirmed that these fast times are critical to prevent system collapse.
Delivery Order Priority	Manager Survey	The Kitchen Order Queue is set to Prioritize based on a numerical label (lbl_OrderPriority). Delivery orders are assigned a higher value at their Start Point, ensuring they are always processed first by available cooks.	To model the critical business rule of prioritizing time-sensitive delivery orders over in-store orders.
On-Site Service Times	Direct Timed Observation ("Undercover" visit)	The Order at Kiosk and Sit At Table activities have their processing times defined by Triangular Distributions based on a sample of 15 observed customers.	To add a layer of empirical, real-world data to customer-facing activities, increasing the model's fidelity beyond survey estimates.
Packaging Costs	Industry Standard	A Finance cost of £0.10 per unit was added to the Hand out activity for delivery orders.	To incorporate variable costs into the model's profitability analysis, allowing for a more accurate financial comparison between different operational scenarios.
Profit Per Meal	Industry Standard	Profit per meals were added using the industry standard	To add a simple business layer for profit analysis

Section 2: Originality and Quality of the Simulation Model

3.1 Model Overview and Design Philosophy:

The simulation model was developed as a dynamic "digital twin" of a high-volume Quick-Service Restaurant (QSR). The primary design philosophy was to move beyond a simple linear queue-and-server model and instead capture the complex, parallel, and often conflicting processes that define modern fast-food operations. The model is structured to trace the complete customer journey through **four distinct modules: Order Placement, Kitchen Production, Final Assembly & Collection, and In-Store Experience (Seating).**

The core of the model's functionality lies in its ability to simulate key system constraints, including staff availability (resources), equipment capacity (replicated Work Centers), and physical space (table availability). This level of detail was deemed appropriate and necessary to accurately investigate the project's key objectives, such as the impact of shifting bottlenecks and menu complexity on overall profitability and customer satisfaction. It was crucial to keep the modeled grounded in reality during this process, to ensure that model's accuracy. To achieve this proved to be challenging as restaurants are complex and include many internal operations that are interchangeable many of the times. Parallel processes such as 1 cook handling many meals at the same time had to be accounted for as well.

To explain how I approached the design, I will go through these four stages to simplify the explanation.

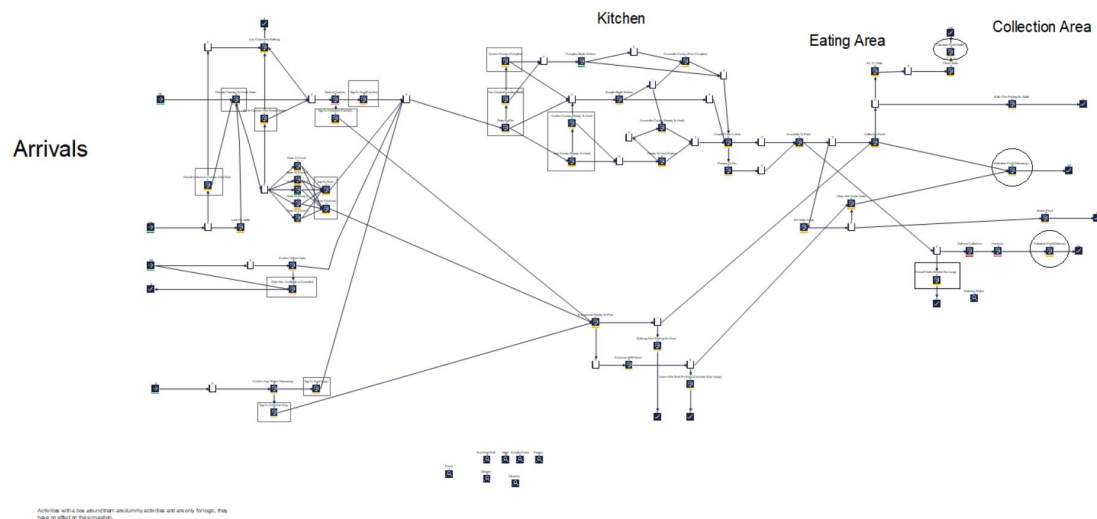


Figure 3.1: Overview of the KFC Queen Street Simulation Model

(Activities with boxes around them are dummy logic activities only)

(Activities with circle around them are for calculating profit using Visual Logic from a spreadsheet)

3.2 The Customer Journey: A Process-Based Description

3.2.1 Customer Arrival and Order Placement: The model simulates four distinct customer arrival channels: Eat-In, Takeaway, Delivery, and App-based Pre-orders.

- **Functionality:** To reflect real-world demand, arrivals are governed by a master Time Dependent Distribution, which modifies the arrival rate based on validated "Normal Hour" and "Busy Hour" data.

- **SIMUL8 Functions:** This section utilizes multiple Start Points, each with its own data-driven arrival logic. Customer choice between the Kiosk (Modeled as separate copies) and the Cashier (a single activity) is handled by probabilistic routing. Similar to the standard structure of QSRs and the modeled restaurant (KFC) the front end staff is flexible and their tasks include cashiering. A label (Order_ID) set to unique is assigned to each coming arrival to stamp each with an order that helps tracks, batch and then collect them at different points through the journey of the work item (With online orders having an additional unique ID as well, helping analyzing the difference between the channels in input and output). To properly create a digital twin for a fast food joint that accurately simulate the operations. Probability distribution was used to associate each work item with a menu item to be ordered at the subsequent ordering work center.

Delivery arrivals were giving priority, with their own arrival pattern. In addition, the delivery orders stop at 11:30. To model scenarios in which we extend the time in which the restaurant accepts online orders, I made it so that at 11:30 the orders will stop coming due to the Delivery riders resource = 0 by using shifts at the delivery riders resource. This directly shows us the amount of online orders that the restaurant could have processed in that additional time.

Order in app before pick up, was also included, to test various scenarios.

3.2.2 The Kitchen Workflow: A Split/Assemble Process:

A core challenge when modeling a restaurant in Simul8 was how to approach the menu variety for profit and cost analysis in restaurant without losing an acceptable level of abstraction. The key factor in determining the approach was if it's capable of answering the following questions:

- 1- What is the difference in time between finishing a complex meal (That might require more cooking time than the usual at a fast food joint, by using different kitchen appliances, with different costs) and a simple meal (That requires few minutes) in the same order, thereby the simple meal might turn cold leading to customer dissatisfaction?

- 2- What's the impact of the drinks station (Where the packer prepares the drinks for the meals) on the system's throughput, queuing times and staff utilization, therefore the impact on the profitability.

- 3 – How does packing a drink affect the waiting times for collection for customers?

- 4- The utilization of cooks and the profitability of each station as well as the costs.

- 5- Crucially, if we move an item from a complex meal category to the ready to heat station (Such as a vegan burger) what is the impact on the restaurant in throughput, profit, costs, staff utilization, queuing times and other data we can extract from the model.

The goal was to answer these questions and some more without going into details about the menu and the items which will complicates the analysis and does not provide us many insights about these problems.

To solve this problem I followed a novel approach, I split the most ordered meals of the restaurant based on collecting data into **3 categories**:

1- Complex Meal Station. 2- Simple Meal Station. 3- Ready To Heat Station.

I then added a simple probability distribution based on menu items + customer's data to decide if the meal has a drink or not, and if it is, then it's sent for the packer to pack a drink with it before sending it to the final assembly point.

To Analyze if a simple meal is waiting too long for complex meal, I simply split the order based on a spreadsheet into either a combo or non combo meal, being that the cook working on the complex meal usually finishes the simple meal and sits it aside or another cook does that, then the simple meal will usually wait for the complex meal in a queue, waiting to be collected and then sent to the Assembly queue to check if it has drink too or not.

By applying this novel solution, the five problems above were solved, allowing us to extract the insights we need from the model, without complicating it too much with unnecessary menu items analysis, which is out of the scope of what we are trying to solve.

Adding the Ready To heat station allows us to find out what happens if we move items from the complex meal station to the ready to heat station. The ready to heat station also follows the same logic of the combos as the complex meals in relation to the simple meal station.

This solution, ensures that the process is also similar to KFC restaurant and similar fast food joints, where the cooks usually are working on multiple items at the same time, therefore, it's logical that a simple meal that enters a kitchen (Containing wings and French Fries) won't take 2 minutes (1 minutes each for example) but actually will take 1 minutes only handled by 1 cook, being that both items are being cooked at the same time. This ensures real world constraints and logistics are being counted for, and grounds our model in how it approaches the solution, ensuring actionable insights for restaurant owners.

Since a A simple queue leading to a single "Kitchen" activity would fail to capture the nuances of a any restaurant, let alone a fast QSR, a custom split-and-reassemble logic was designed using an Order Splitter activity. This dummy Work Center (0 processing time) uses **Label-based Routing and spreadsheets** to separate combo meals into their constituent parts: "Simple" items (e.g., fries, drinks) and "Complex" items (e.g., burgers) or "Ready to Heat" items (like hot desserts that needs microwaving).

Model Detail: The three meal types are processed in parallel, being an order is composed of 1 simple item and 1 complex or ready to heat item, this allows us to extract insights about the various meal types without getting into too much details about the menu items:

- Simple Meals Station: Modeled using Replicated Work Centers to represent multiple, identical fryers.
- Complex Meals: Modeled as a single Work Center that can be operated by Multiple Resources (Specialized Cooks) simultaneously, representing a shared grill/assembly area. This distinction in modeling approach was critical for accurate bottleneck analysis.
- Ready To Heat Station: Modeled as a single Work Center that can be operated by a simple cook, this represents appliances such as a microwave.

Functionality: After processing, items are sent to Assemble Combo activities, which use the Collect discipline to ensure all parts of an order are present before it is considered complete.

Finally, this approach ensures that not all orders have to be combos, some orders just go to their station directly (like wings only, burger only with no meal and so on).

3.2.3 Final Assembly, Collection, and Customer Patience:

Functionality: Completed meals wait in a **Customer Waiting Area queue**. This queue is serviced by a **Packer resource**. The model uses the Shelf Life feature on this queue to simulate customer "balking." If an order's waiting time exceeds **25 minutes**, it is routed to a **Wasted Food End Point**, and the **customer leaves**, representing a direct loss of revenue and a key performance indicator. In addition, a customer might go away and then come back later, this is accounted using the data by the survey and when a customer is not ready to pick up the order when it's shouted by the cashier then the customer will leave to the "Away Customer Zone" queue and then proceed to be "absent" of an average time derived from the survey till coming back and then claiming their order or in some cases coming back and finding that their order was sent to waste (after 25 minutes).

The orders are collected in the collection area by the Order ID label of both the customer and the meal, only when the meal has finished and combined or not with a drink.

For the drinks, "Check if has a drink" requires a quick few seconds check from a packer to see if there's a drink with the meal or not, if yes then the packer prepares the drinks. This directly lets us measure utilization and staff's roles variation inside the kitchen.

3.2.4 The Dine-In Experience: Table Management:

A novel approach : To analyze the critical constraint of physical space, a dedicated logic module for table management was created. Customers who select "Eat-In" are routed to a Look For Table process. If no tables (modeled as a Resource) are available, customers enter a queue (Wait for Table). This queue also utilizes Shelf Life to model balking and the diversion of customers from "Eat-In" to "Takeaway," providing a direct measure of revenue lost due to insufficient seating. This is different than the customers who bulk after collecting their items and leaving at the Eating Area. This allows us to analyze the difference between a possible automatic table allocation system and the chaotic current one where customers might come in, find a table empty, then when ordering and collecting their food, their items might enter the complex kitchen station and a later customer who arrives and orders a simple meal from the simple kitchen station will also find the same table and sits on it. The earlier customer then will collect their food and find "their table" gone, taken by the later customer.

The process look for table requires a table resource then releases it. While Sit at table requires the resource but does not release it. To find the difference between automatic allocation at order time and between first come first serve basis system, we simply change the the resource of Tables at "Look For Table" activity from require and release here to require here only.

3.3 Input and Output Variables

3.3.1 Key Input Variables (Model Flexibility):

The model is highly data-driven, allowing for a wide range of "what-if" scenario analysis by modifying parameters primarily stored in spreadsheets within the Information Store in few seconds.

Input Variable	SIMUL8 Object / Feature	Example Use
Demand Patterns	Time Dependent Distribution	Adjusting for weekend vs. weekday demand.
Staffing Levels	Resource Shifts	Testing the impact of adding a cook or packer during peak hours.
Menu Mix & Process	ss_KitchenRouting Spreadsheet	Re-routing the Vegan Burger from "Complex" to "Ready To Heat", in addition to making a menu item a combo or not by simply changing the spreadsheet combo label from 0 to 1.
Service Times and routing	Activity Timing Distributions	Modeling the impact of a new, faster oven.Or removal of kiosks.

3.3.2 Key Output Variables (Key Performance Indicators):

The model is designed to capture a balanced scorecard of performance, measuring efficiency, financial success, and customer experience.

KPI Category	Output Variable	SIMUL8 Source
Throughput	Total Orders Completed (per channel)	Number Completed at End Points
Financial	Total Profit, Total Costs (by resource)	Results Manager (from Finance feature)
Customer Experience	Average Time in System, Average Wait Time before collection, etc	lbl_TotalWaitTime, Queue Results
Service Level	% of Orders Collected < 10 mins, # of Balking Customers	Queuing Time within limit, End Point Total Entered
Utilization	Staff & Equipment Utilization %	Resource and activities Results

3.4 Summary: Model Quality and Appropriateness

The complexity of this simulation model is a direct and necessary reflection of the system it represents. A simpler model would have failed to identify the critical insights of this study, such as the "shifting bottleneck" between the kitchen and packing stations, the "cold food" effect of mismatched processing times, and the significant revenue loss caused by a lack of available tables. By employing a wide range of SIMUL8's advanced features—from pooled resources (For Example the Cleaners pool resource uses front staff and cleaner resources, being that the cleaner resource is available only in busy hours) and time-dependent distributions to data-driven logic via spreadsheets—the model successfully functions as a robust and validated tool for deriving general principles for operational improvement in the QSR industry.

Section 3: Simulation Experiments and Results

For the final analysis and verification phase, I followed an important design rule: First validate the model to the real world data, and once verified that the arrival rate and the amount of orders of the dataset leads to the same amount of orders at the end of the day as in the model with the same arrival rate, being that the processes and activities between arrivals and end points follow the datasets time inputs, then the model is ready for trials.

4.1 Validation and Verification phase:

The core philosophy of this work was to first construct and rigorously validate a "digital twin" of a Quick-Service Restaurant (QSR), and only then to use this trusted model as a predictive tool for "what-if" scenario analysis. This two-stage approach ensures that the insights derived from the experiments are grounded in a credible and realistic representation of the actual system.

The validation process focused on ensuring the model's outputs, under a specific set of inputs, accurately mirrored the real-world performance metrics provided by primary and secondary data sources. Once this baseline accuracy was established, a series of structured experiments were conducted to explore the key objectives of this study.

The primary validation metric was the model's ability to replicate the total daily order volume as estimated from the survey data. The manager and google maps busy hours indicated a typical weekday consists of approximately 3 peak "Busy Hours" (at ~70 orders/hour) and 11 "Normal Hours" (averaging ~25-30 orders/hour, for which we used an average of 28).

- Normal Hour Volume: 11 hours * 28 orders/hour = 308 orders
- Busy Hour Volume: 3 hours * 70 orders/hour = 210 orders
- Theoretical Total Daily Orders: 308 + 210 = 518 orders

This gives us about 518 orders approx per day. These hourly rates were used to parameterize the model's master Time Dependent Distribution for customer arrivals, which was based on google map's data and the survey. Furthermore, the other activities including back and front of the kitchen were also parameterized using the data sources outlined in the data section. Once all the data was input, the model was then run for a verification. The goal was to ensure that the model outputs the same estimated daily orders by the manager in the survey, to ensure accuracy of the model.

When the simulation was executed for a full 14-hour day with these parameters, the model's output was 503 total completed orders. This represents a variance of only **2.9%** from the theoretical target derived from the manager's data.

Furthermore, the model predicted about 4 to 11 customers would not order or receive their orders due to a lack of available tables or long queues for collection. This quantitative output aligns perfectly with the manager's qualitative real-world estimate of "3-10 people" leaving for the same reasons on a busy day.

**Model Validation: Simulated vs. Theoretical Daily Order Volume
(Variance: -2.9%)**

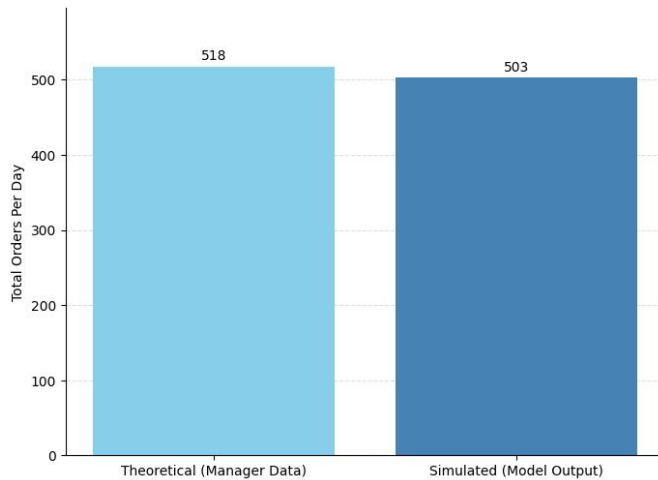


Figure 4.1: Comparison of theoretical daily order volume calculated from manager data against the simulation model's output, showing a variance of less than 3%, thus validating the model's predictive accuracy

4.2 Experiments And Trials: Now that we verified the model is ready, being that every activity when matched in data input with its other twin's produces the same output, we can conduct various trials, scenarios and experiments to gain insights about the operations of a Quick service restaurant.

4.2.1 Experiment 1: Identifying the True System Bottleneck (Sensitivity Analysis): An initial hypothesis was that the kitchen's processing time was the primary constraint. This experiment was designed to test the system's sensitivity to this variable.

- **Baseline Scenario:** Utilized the manager's validated, high-speed cook times (0.5-1 minute for complex meals).
- **"Stressed System" Scenario:** Utilized a more conservative, longer cook time of (5 minutes average per complex meal) to simulate a less efficient operation in competitive restaurants with more complex meals that require an oven (Such as McDonald's). This was taken from ex-employees' testimonials from forums like reddit and quora.

The results of this sensitivity analysis were stark and conclusive. While the baseline scenario performed optimally, the "Stressed System" scenario, with a modest increase in kitchen cycle time, resulted in a catastrophic failure of the entire operation.

- **Service Level Collapse:** The number of customers who left without collecting their food (balked due to excessive waits) skyrocketed from a near-zero baseline to 259 customers. This represents a complete breakdown in the restaurant's ability to serve its waiting customers. Queuing time within limit for waiting for collection queue (customers waiting to collect) was set for 10 minutes, only 39% of ordered collected, with the average waiting time being 16.47 minutes and the maximum being 25 minutes before customers leave out of frustration.
- **Online Order Fulfillment Failure:** The system's capacity to handle online orders was severely compromised, with only 98 out of 133 (74%) of online orders being completed. The remaining 35 orders were effectively lost, representing a direct hit to a key revenue channel.

Insight: This experiment definitively proves that the entire QSR system is balanced on a "razor's edge" of kitchen efficiency. The data demonstrates that the high-speed, sub-minute processing times reported by the manager are not just an operational target but a critical success factor essential for the restaurant's survival. A seemingly small degradation in kitchen speed does not lead to a small decrease in performance; instead, it triggers a cascading system failure that results in massive customer loss and a collapse in throughput. This validates the intense focus QSRs place on process standardization and speed within their kitchens, and must rely on pre prepared meals to be cooked in under a minute in busy hours.

4.2.2 Experiment 2:

- **Rationale:** The validated baseline model identified physical space as a significant constraint on revenue. On a weekly basis, an average of 12 customers were lost due to a lack of available seating. This occurred in two distinct ways: 2 customers balked upon arrival (pre-order), while a further 10 customers, having already collected their food, were unable to find a table and subsequently left ("post-collection balking"). This experiment was designed to test two hypotheses: first, whether an automated table allocation system could mitigate these losses; and second, how sensitive the system is to a reduction in its physical seating capacity—a common scenario for smaller, high-traffic urban locations. **To conduct** this sensitivity analysis, the number of available Table resources was reduced from the manager's stated 30 to a more constrained 10 tables. Two scenarios were then run:

Scenario 1 (Baseline: First-Come, First-Served): In this configuration, "Eat-In" customers only attempt to secure a Table resource *after* collecting their food from the Collection Point. The Table resource is required and occupied only during the final "Sit At Table" activity. This models the current chaotic, FCFS system.

Scenario 2 (Automated Allocation): To simulate a "reserve-at-order" system, the model's logic was fundamentally altered. The "Look For Table" activity was moved to occur *before* the customer's order is sent to the kitchen. Crucially, the Table resource was set to be Required at this initial stage but Not Released until after the "Clean Table" activity was complete. This effectively reserves the physical table for the entire duration of the customer's order and collection process.

Key Findings: The simulation results clearly demonstrate that under conditions of physical constraint, the proposed "Automated Allocation" system is catastrophically inefficient.

Under the baseline FCFS system with 10 tables, the number of balking customers increased significantly, rising to 240 per week. However, in the "Automated Allocation" scenario, this number surged further to 260 per week.

Analysis of the simulation trace reveals the cause: a severe reduction in table turnover. By reserving a table at the point of order, the valuable Table resource is occupied for the entire duration of the customer's wait for food—a period during which it provides no value and is effectively "blocked." In the FCFS system, the process of finding a table is decoupled from food preparation,

allowing both to run in parallel. This "breathing room" ensures tables are utilized only for the value-added activity of eating, leading to a much more efficient flow, especially when seating is limited.

Conclusion: This experiment provides a powerful insight for QSRs operating in space-constrained environments. A seemingly chaotic first-come, first-served seating policy is substantially more efficient than a structured "reserve-at-order" system. The proposed automated system, while attractive in theory, would lead to a significant loss of revenue by crippling the utilization of the restaurant's most limited physical asset: its seating space. It proves that for processes with significant in-kitchen lead times, decoupling parallel activities is a more robust strategy than creating rigid dependencies.

4.2.3 Experiment 3, Menu Engineering Analysis:

A key application of this validated QSR model is its use as a tool for Menu Engineering—the strategic analysis of how changes to a menu's composition and production methods impact system-wide performance and profitability. This experiment explores this capability in two parts. A financial layer was integrated into the model, assigning a Profit Per Unit to each menu item within a dedicated spreadsheet (ss_KitchenRouting) and Usage Costs to key Work Centers. While these figures are estimates for analytical purposes, they provide a robust framework for comparing the relative profitability of different operational strategies.

Part A: To test the hypothesis that moving a menu item to a more "efficient" (faster or cheaper) production method will always lead to an increase in overall profit. This scenario is designed to explore the risk of unintended consequences in a resource-constrained system.

Scenario Setup: To model an operation with specialized equipment, a Specialized Cook resource was created and assigned to the Complex Meals Station, which also carried a higher operational Usage Cost. In the Baseline Scenario, the "Vegan Burger" is defined as a complex meal. In the "Re-engineering" Scenario, its routing is changed via the ss_KitchenRouting spreadsheet to the less costly Ready-To-Heat Station. To reflect the higher cost of pre-packaged goods, its Profit Per Unit was reduced from £4 to £2 for this scenario. **Baseline Scenario:** The "Vegan Burger" is defined in ss_KitchenRouting as a "Complex Meal," requiring the Specialized Cook, that takes more cooking time to cook, as he prepares it from a dry mix. **"Re-engineering" Scenario:** The ss_KitchenRouting spreadsheet is modified to re-categorize the "Vegan Burger" as a "Ready-to-Heat" item, shifting its production to the less costly Ready-To-Heat Station which only requires a Simple Cook, this is done by simply changing the station. We decrease its profit slightly as it costs more to pre package it and store it than a dry mix.

Key Findings: The simulation produced a counter-intuitive financial result. Despite the more efficient processing, the overall daily profit increased by a negligible £3. Analysis of the Resource Utilization and Throughput Results reveals the cause: a shift in the product mix. By re-routing the Vegan Burger, the Specialized Cook resource experienced more idle time. If this resource is pooled to assist with other tasks, it can lead to an increase in the production of lower-margin simple meals at the expense of higher-margin complex meals. This demonstrates the principle of "resource cannibalization" where a localized gain can lead to a less profitable overall product mix if not managed carefully. A £3 increase does not justify the loss of customers who prefer hand made dry mixes.

Part B: Stress-Testing for Marketing Promotions (KFC-Specific Application): We evaluate the kitchen's capacity to handle a marketing promotion that skews the product mix towards a resource-intensive item, and identify the system's breaking point.

Scenario Setup: A "Family Feast" promotion was simulated by altering the Time Dependent Distribution for customer arrivals for a single day. The probabilities within the dist_OrderChoice distribution were adjusted to heavily favor the "Chicken Bucket" (a complex meal) by 85 % over all other items.

Key Findings: The model predicted that the promotion, while increasing potential revenue, would overwhelm the system's current capacity. The number of customers balking from the collection queue rose to 27. A second trial was run where a fifth cook was added. This resulted in only a minor profit increase of ~£100 and a reduction in cook utilization from 71% to 57%, without significantly reducing the number of balking customers.

This indicates that the cooks were not the primary bottleneck. A deeper analysis of the Queue Results for the Customer Waiting Area (the queue before packing) revealed that this is where the new bottleneck emerged. The one point during the afternoon, 76 people were waiting for their food to be collected, by the busy front end staff and packers. Since the packers were busy with the handing off, they couldn't keep with the drinks, as they focused on the packing of chicken buckets. The kitchen was able to produce the buckets, but the packing station was unable to keep up with the surge. This demonstrates a classic shifting bottleneck: improving one part of the system (cooking capacity) simply exposed the next weakest link (packing). The insight is not that more cooks are ineffective, but that any successful promotion must be supported by a holistic analysis of its impact on the entire process chain, from the kitchen to the final hand-off.

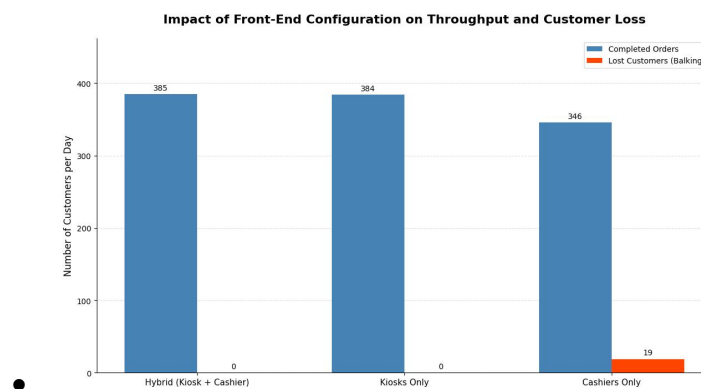
4.2.4: General Insights:

1-While self-service kiosks are now standard in the UK QSR market, many international markets still rely on a traditional cashier-only model. This experiment was designed to quantify the operational and financial impact of different front-end configurations to determine the optimal strategy for maximizing throughput while managing costs. We test three scenarios, Hybrid Model where customers can choose between each with 80% choosing a kiosk, reflecting real world. Kiosk-Only Mode where 100% of the customers use a kiosk, and then a Cashier-Only Mode.

The results demonstrate a stark difference in performance between the models, providing clear, data-driven insights.

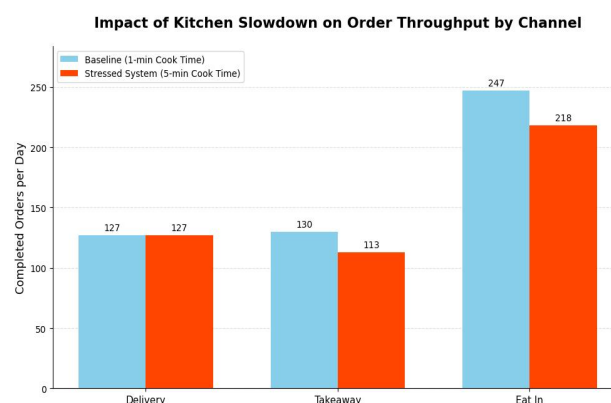
The Cashier-Only scenario resulted in a systemic failure. The serial processing nature of the cashiers was unable to handle the arrival volume, leading to an average customer wait time exceeding 25 minutes and causing 19 customers to balk from the queue—a direct loss of revenue. This finding proves that the throughput levels of a modern, high-volume QSR cannot be sustained with a traditional cashier-only system, even when fully staffed.

Furthermore, a comparison between the Hybrid Model (385 completed orders) and the Kiosk-Only Model (384 completed orders) revealed a negligible difference in total throughput (less than 1%). This is a critical insight: it demonstrates that the capital investment in kiosks effectively handles the vast majority of order volume. The cashier's role in order-taking becomes largely redundant, suggesting that removing this position would not negatively impact throughput, thus offering a significant operational cost saving.



2-When the kitchen is under heavy pressure (5 minutes average per order), the model revealed a critical insight into the restaurant's operational priorities. While overall throughput collapsed, the impact was not distributed evenly across all order channels. **Delivery** orders demonstrated high resilience, with the completed orders staying the same. **This** is a direct result of the "Delivery Priority" rule implemented in KFC. In-store channels (**Takeaway and Eat-In**) absorbed the vast majority of the negative impact. Takeaway orders, for example, plummeted by 49, as these customers were forced to wait and ultimately barked.

This demonstrates that while prioritizing deliveries protects a key revenue stream, it does so by sacrificing the in-store customer experience during periods of operational stress. It proves that modern, multi-channel QSRs are highly sensitive systems, where even minor degradations in kitchen efficiency can lead to a significant and disproportionate decline in service for non-priority channels.



Section 4: Conclusion

This project successfully developed and validated a highly-flexible discrete-event simulation model of a typical Quick-Service Restaurant, using a KFC franchise as a real-world baseline. The primary objective—to create a "digital laboratory" for deriving general principles for operational improvement—was achieved by moving beyond a simple linear model to capture the complex, parallel, and resource-constrained nature of modern QSR operations. The scope was reduced to an archetype that is similar in its operational approach and methods, this allows it to be flexible while staying within constraints.

The validation phase confirmed the model's accuracy, with its outputs for daily throughput and customer balking rates aligning within a 3% variance of manager-provided data. Subsequent experiments yielded several critical, and at times counter-intuitive, insights:

1-System Fragility: The simulation proved that the entire QSR system operates on a "razor's edge" of kitchen efficiency. A minor degradation in the sub-minute cook times for complex meals was shown to trigger a catastrophic collapse in service levels, confirming that speed in the kitchen is the most critical operational factor

2-The Shifting Bottleneck: It was demonstrated that improving one part of the system (e.g., adding more cooks) simply exposes the next weakest link. In this model, the bottleneck consistently shifted from the kitchen to the final packing and collection stage, proving that holistic, system-wide analysis is essential before investing in resource changes.

3-The hidden side of "novel" Solutions: The model showed that seemingly logical improvements, such as an automated table reservation system or re-routing a menu item to a faster production line, can have negative consequences. These changes were found to cripple table turnover and negatively impact the overall product mix, leading to a significant decrease in profits. Initially I thought these were obvious solutions but the model proved otherwise.

Ultimately, this study demonstrates the immense value of simulation as a predictive, data-driven decision-making tool. By providing a risk-free environment to test everything from staffing strategies to marketing promotions and new technology, the model successfully quantified the impact of operational decisions on both the customer experience and the financial bottom line. The principles derived from this model—regarding the management of bottlenecks, the importance of staff flexibility, and the hidden costs of physical constraints—are applicable across the broader QSR industry, offering a robust framework for continuous improvement. I have learned through this the importance of simulation as a realistic digital twin that reveals the truth about assumptions and how in reality a lot of what is expected is not actually what happens in ground.

Section 5: References

Curin, S. A., Vosko, J. S., Chan, E. W., & Tsimhoni, O. (2005). *Reducing Service Time at a Busy Fast Food Restaurant on Campus*. Proceedings of the 2005 Winter Simulation Conference.

Online forums such as reddit and Quora for ex employees provided data.

Appendix:

Appendix A: Data Collection Instruments & Raw Data

• A.1: Manager Survey Instrument:

Cardiff University Restaurant Survey

1 What percentage of orders are Eat-in, Delivery, and Takeaway? : Eat in: 50% Delivery: 25% Takeaway: 25%
2 How many orders you get in a busy hour, and quite hour (average): Busy hour: 70 Normal Hour: 25-30
3 Your 3 Most selling meals : 1: Burgers meals 2: Chicken Buckets 3: Wraps
4 In a busy hour how long does it take (minutes) for the kitchen (average) to cook : A Complex Meal (big variety special wrap): 30sec-1min A simple order(burger): 30sec
5 Do Cooks Prepare in Advance some meals to save time (do you pre-make popular burgers or have drinks ready to go?): Yes
6 If a customer not available to pick their food, how long you wait (minutes) before you cancel the order : 5min-25min Percentage of customers not picking up order immediately: 1-2%
7 Do you waste the meal no one picks or serve it other customers (Answer One Word (Serve) or (Waste): Yes/Waste
8 In Normal hour how many staff are : Cashiers(Front Staff): 2 Kitchen(Cooking): 1 Packing Orders(Packers): 2 Lobby or Cleaner: 0
9 In Busy Hour how many staff are : Cashiers(Front Staff): 4 Kitchen(Cooking): 2 Packing Orders(Packers): 3 Lobby or Cleaner: 1
10 On a busy day, roughly how many customers do you think leave without ordering because the queue is too long : 3-10 people
11 Finally, when a delivery order arrives do you have priority for the order over in store (Answer Below yes or no, or it depends):
Yes: ☒
No: ☐
It depends on: ☐

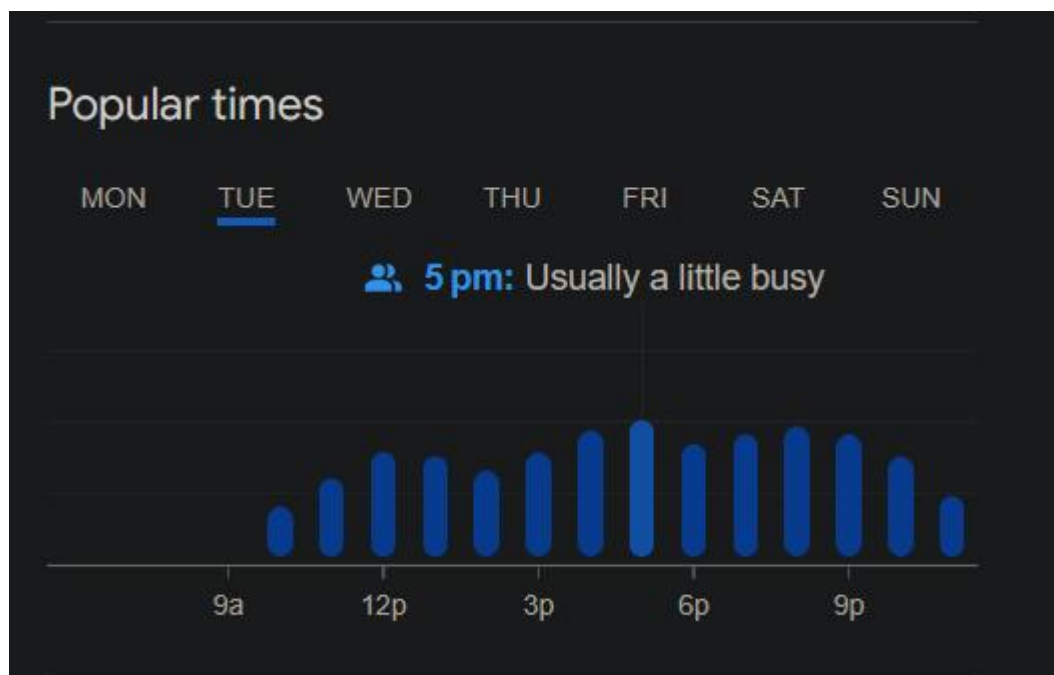
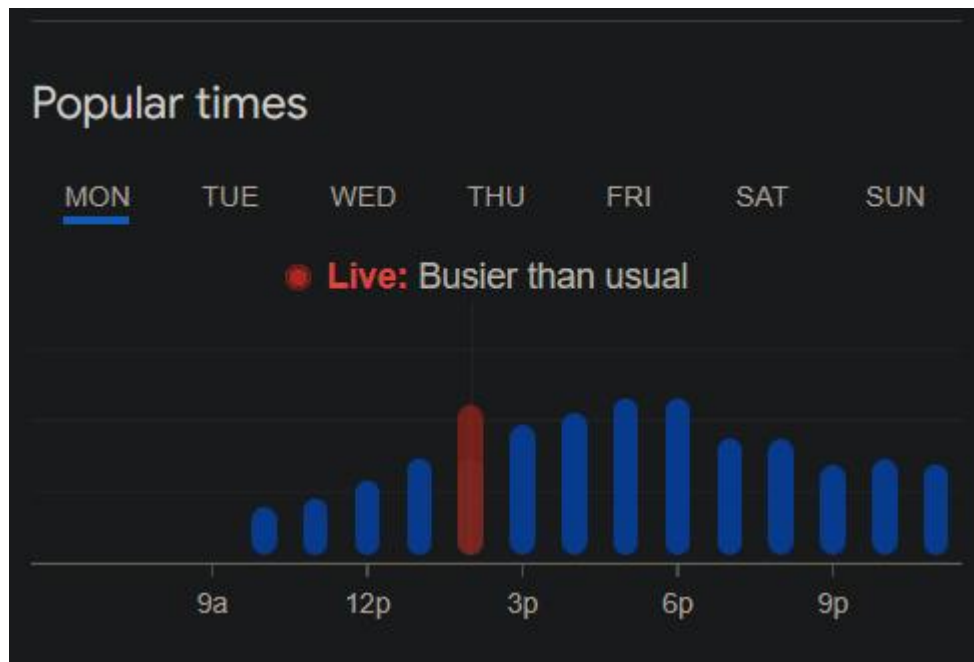
Thank you very much for your help!

- **A.2: Direct Observational Data:**

Customer Interaction and Occupancy Data

Customer	Interaction Type	Interaction Time (min)	Table Occupancy (min)
1	Kiosk	1.3	25
2	Kiosk	1.95	30
3	Kiosk	1.6	22
4	Kiosk	1.8	35
5	Kiosk	3.5	28
6	Kiosk	1.4	20
7	Kiosk	3.2	32
8	Kiosk	1.7	26
9	Kiosk	1.65	24
10	Kiosk	1.85	38
11	Kiosk	1.55	23
12	Kiosk	1.2	40
13	Cashier	3.0	18
14	Cashier	2.5	42
15	Cashier	3.7	15
Averages	Kiosk Interaction	1.89	
	Cashier Interaction	3.07	
	Table Occupancy		27.87

- **A.3: Secondary Data (Google Maps):**



Appendix B: Model Parameterization & Logic

- B.1: Master Data Spreadsheet (ss_KitchenRouting):

The screenshot shows a Microsoft Excel spreadsheet titled 'Sheet: ss_KitchenRouting'. The spreadsheet contains data for various meal items, including their station IDs, names, and profit calculations. The data is organized into columns: A (Meal Type), B (Station ID), C (Meal Name), D (lbl_IsCombo), E (Profit Per Unit), F (Total Profit Per Item (Eat In)), G (Total Profit Per Item (Takeaway)), H (Total Profit Per Item (Delivery)), and I (Total Profit Per All Items). The data rows show items like Vegan Burger, Burger Meal, Pudding, Chicken Drumsticks X3, Vegan Combo With Fries, Chicken Buckets, Hot Chicken Salad & Fries, and Wings, with their respective profit values.

	A	B	C	D	E	F	G	H	I	J
	Meal Type	Station ID	Meal Name	lbl_IsCombo	Profit Per Unit	Total Profit Per Item (Eat In)	Total Profit Per Item (Takeaway)	Total Profit Per Item (Delivery)	Total Profit Per All Items	
1										
2	1	1	1 Vegan Burger	0	4	160	72	72	7738	
3	2	1	1 Burger Meal	1	3	93	66	36		
4	3	3	3 Pudding	0	4	352	176	176		
5	4	2	2 Chicken Drumsticks X3	0	2	36	14	24		
6	5	1	1 Vegan Combo With Fries	1	4	88	32	40		
7	6	1	1 Chicken Buckets	0	2	3004	1554	1448		
8	7	3	3 Hot Chicken Salad & Fries	1	5	135	80	80		
9	8	2	2 Wings	0	3	0	0	0		
10						0	0	0		
11						0	0	0		
12						0	0	0		
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34						0	0	0		

- B.2: Key Label Definitions:

Label Name	Description & Use
lbl_OrderPriority	A numerical label set at the Start Point to ensure Delivery orders are processed first in the kitchen queue
lbl_SimpleMealReadyTime	A "stopwatch" label used to calculate the time a simple meal waits for a complex meal during assembly, a key metric for food quality
Lbl_order_ID	A numerical label set to Unique to give each customer his own order Id and is used to collect the meals parts and customer and his meal
Lbl_ArrivalTime	Tracks the arrival times for customers and orders, important for debugging and extracting insights about the work item journey
Has_drink	Uses dist_drinks for a chance the meal has a drink or not
Lbl_MenuItem	Uses dist_OrderChoice which is a probability distribution with a number equal to the spreadsheets menu Items. The label connects the items in the spreadsheets with each order using a probability distribution.
Lbl_dineOrTakeAway	Set to a fixed value representing the order channel, so that the assembly activity then knows where to send the item after its finished from the kitchen

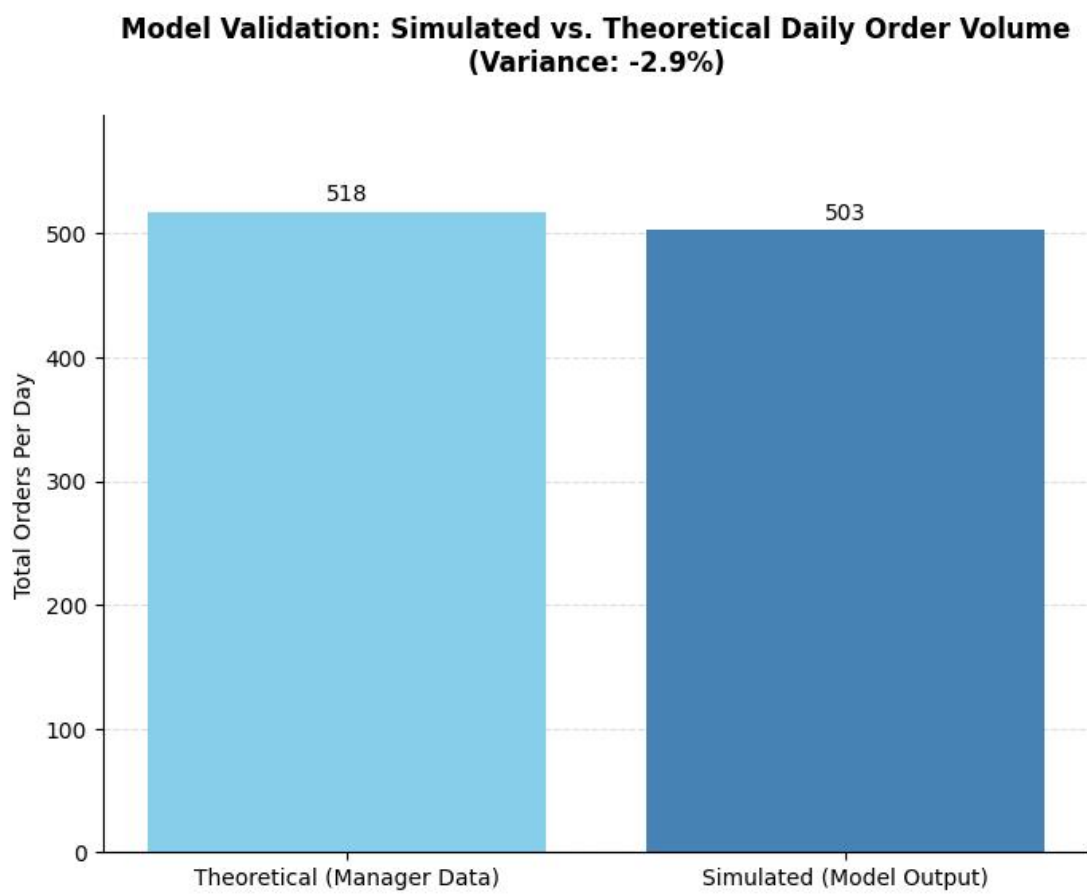
Lbl_collectionType	Fixed value for the collection type at the collection point in store, being takeaway or eat in
Order_type	Uses probability distribution for the ordering method in store : Cashier or Kiosk
Online_order_ID	An Id for online orders so they can be easily tracked
Lbl_IsCombo	Sets the meal if it is a combo meal or not, using the spreadsheet
Lbl_menuType	Routes the meal in the Order Splitter activity to the appropriate channel

B.3: Important Visual Logic Functions:

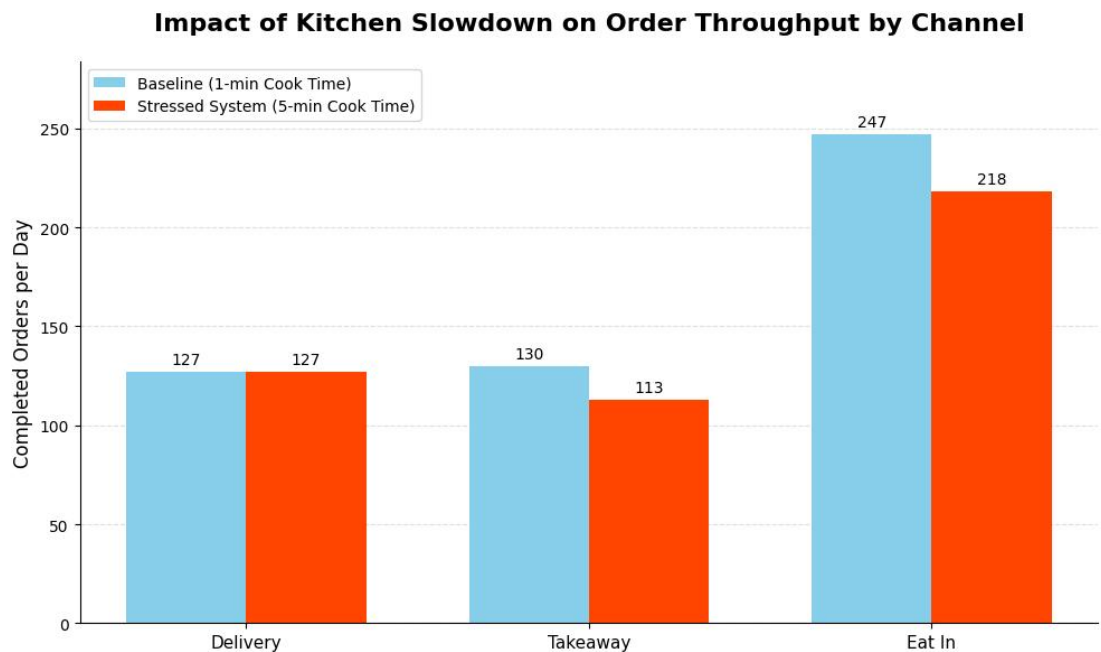
Function Place	Function Description
Kitchen Order Queue - On Exit	Sets lbl_isCombo = ss_KitchenRouting[4,lbl_MenuItem+1]
Kitchen Order Queue - On Exit	Sets lbl_menuType = lbl_ss_KitchenRouting[2,lbl_MenuItem+1]
Complex Meal Station	An optional example setter of dist_CurrentCookTime = ss_CookTimes[2,lbl_MenuItem] It sets the cooking time based on the meal from a spreadsheet.
Calculate Profit Logic Activities	SET gbl_MenuItem = lbl_MenuItem+1 SET gbl_ItemProfit = ss_KitchenRouting[5,gbl_MenuItem] SET ss_KitchenRouting[7,gbl_MenuItem] = ss_KitchenRouting[7,gbl_MenuItem]+gbl_ItemProfit This function sets the menu item profits and writes them to the spreadsheet after each run.
On End Run	A setter logic for key profit values at the spreadsheet
Start Run	A setter logic for key profit values at the spreadsheet

Appendix C: Simulation Experiment Results (Graphs)

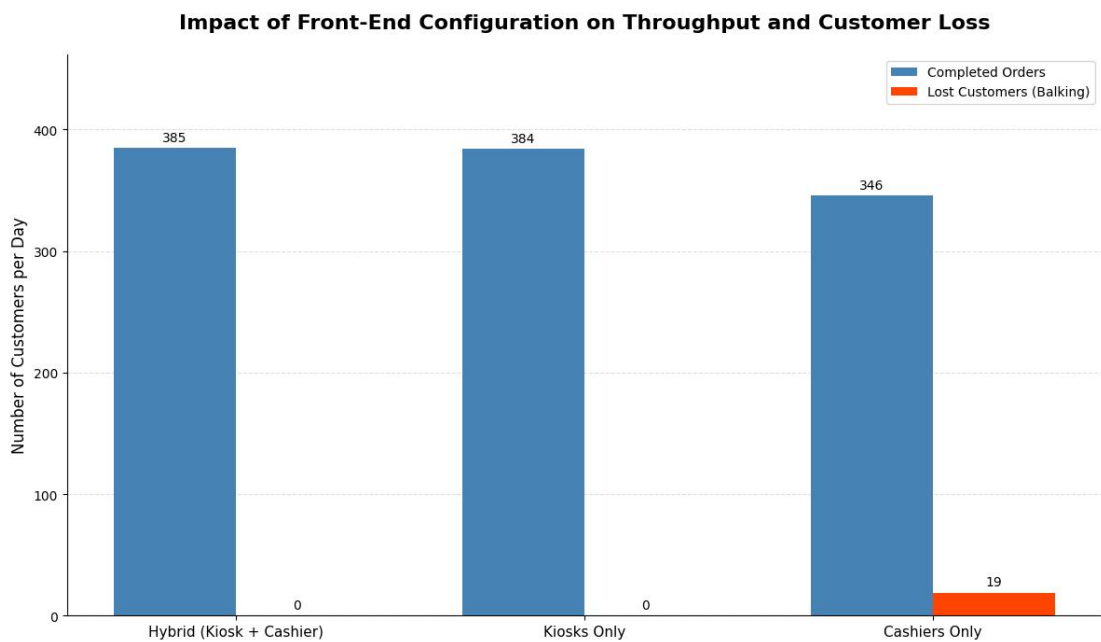
- Figure C.1: Model Validation: Simulated vs. Theoretical Daily Order Volume



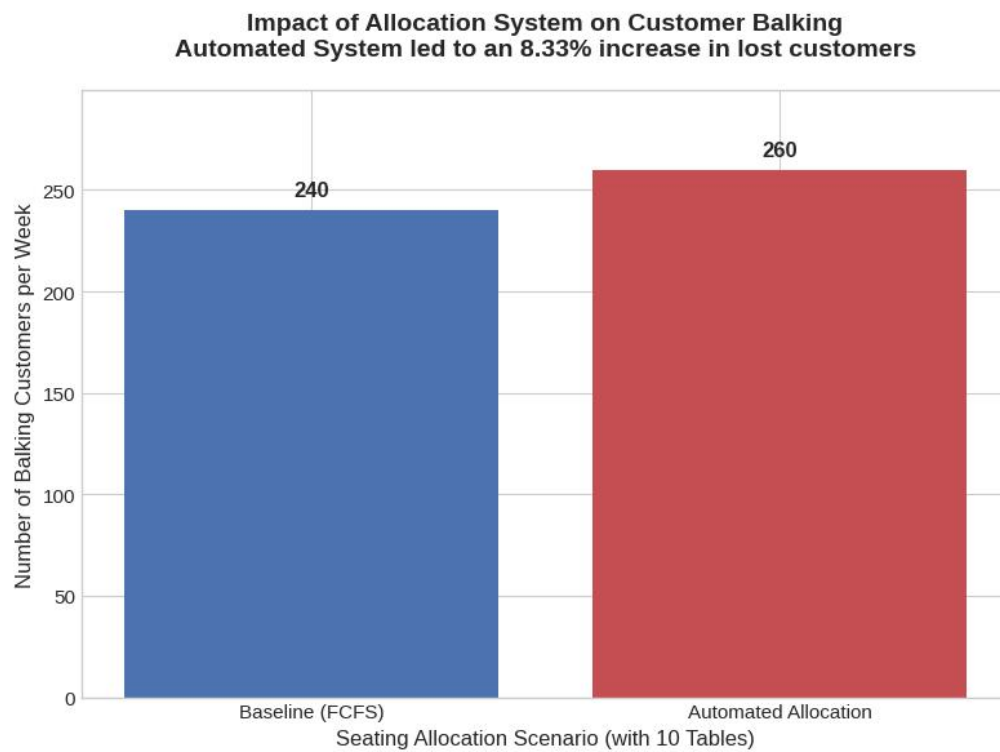
- Figure C.2: Impact of Kitchen Slowdown on Order Throughput by Channel



- Figure C.3: System Throughput Under Different Front-End Configurations (Kiosk vs. Cashier)



- Figure C.4: Effectiveness of Automated Table Allocation on Customer Balking



- Figure C.5, C6: Impact of Staffing Strategies on Collection Point Wait Times

